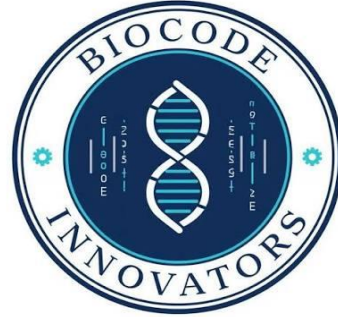




Bioinformatics Bootcamp 2026

Assessment # 03

Sequence Analysis and GC/AT Content Analysis

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Date: 26/07/2026

Q1. Sequence Analysis

What is sequence analysis? Explain its importance in bioinformatics and discuss any four of its major applications.

Answer:

Sequence analysis is the process of examining DNA, RNA, or protein sequences to understand their structure, function, evolution, and biological significance.

Importance in Bioinformatics

Sequence analysis helps researchers identify genes, detect mutations, study evolutionary relationships, and predict biological functions.

Four Major Applications

1. **Gene Identification:** Locates genes and coding regions in DNA sequences.
2. **Mutation Detection:** Identifies genetic variations associated with diseases.
3. **Evolutionary Studies:** Compares sequences to determine evolutionary relationships among organisms.
4. **Drug Discovery and Personalized Medicine:** Helps identify drug targets and develop individualized treatments.

Q2. GC and AT Content

Define GC content and AT content. Why are these parameters important in genome analysis and molecular biology?

Answer:

GC Content

GC content is the percentage of nucleotides in a DNA sequence that are Guanine (G) and Cytosine (C).

$$GC\ Content = \frac{G + C}{Total\ Nucleotides} \times 100$$

AT Content

AT content is the percentage of nucleotides that are Adenine (A) and Thymine (T).

$$AT\ Content = \frac{A + T}{Total\ Nucleotides} \times 100$$

Importance

- Influences DNA stability.
- Helps classify and compare genomes.
- Affects gene expression and DNA replication.
- Useful in evolutionary and molecular biology studies.

Q3. Scientific Problem

The following DNA sequence was obtained from a (Homo sapiens TP53 mRNA).

NM_000546.6 Calculate:

Total sequence length

Number of A, T, G, and C nucleotides

GC Content (%)

AT Content (%) Show your calculations clearly.

Answer:

Given Sequence

Homo sapiens TP53 mRNA (NM_000546.6)

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Homo sapiens tumor protein p53 (TP53), transcript variant 1, mRNA

NCBI Reference Sequence: NM_000546.6

[FASTA](#) [Graphics](#)

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LOCUS	NM_000546	2512 bp	mRNA	linear	PRI 21-NOV-2025
DEFINITION	Homo sapiens tumor protein p53 (TP53), transcript variant 1, mRNA.				
ACCESSION	NM_000546				
VERSION	NM_000546.6				
KEYWORDS	RefSeq; MANE Select.				
SOURCE	Homo sapiens (human)				
ORGANISM	Homo sapiens Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo.				

Homo sapiens tumor protein p53 (TP53), transcript variant 1, mRNA

NCBI Reference Sequence: NM_000546.6

[GenBank](#) [Graphics](#)

>NM_000546.6 Homo sapiens tumor protein p53 (TP53), transcript variant 1, mRNA

```
CTCAAAAGTCTAGAGCCACCGTCCAGGGAGCAGGTAGCTGCTGGGCTCCGGGGACACTTTGCGTTCGGGC
TGGGAGCGTGCTTTCCACGACGGTGACACGCTTCCCTGGATTGGCAGCCAGACTGCCTTCCGGGTCACTG
CCATGGAGGAGCCGACGTACAGATCTAGCGTCGAGCCCCCTCTGAGTCAGGAACATTTTCAGACCTATG
GAAACTACTTCTGAAAAACACGTTCTGTCCCCCTTGCCGTCCCAAGCAATGGATGATTTGATGCTGTCC
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ACACTTTTCGACATAGTGTGGTGGTCCCCATGAGCGCCTGAGGTTGGCTGTGACTGTACCACCATCCA
CTACAACATACATGTGTAAAGTTCCTGTCATGGGCGGCATGAACCGAGGCCATCCTCACCATCATCACA
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CACTAAGCGAGCACTGCCCAACAACACAGCTCTCTCCCGCCAAAGAAAGAAACCACTGGATGGAGAA
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TCAAGGATGCCAGGCTGGGAAGGAGCCAGGGGGGAGCAGGGCTCACTCCAGCCACCTGAAGTCCAAAAA
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CCACTTCTTGTCTCCCACTGACAGCCTCCACCCCCATCTCTCCCTCCCTGCCATTTGGGTTTGGGT
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TCCACTGAACAAGTTGGCTGCACTGGTGTCTTTGTTGTGGGGAGGAGGATGGGGAGTAGGACATACCAGC
TTAGATTTTAAGGTTTTTACTGTGAGGGATGTTTGGGAGATGAAGAAATGTTCTTGCAGTTAAGGGTTA
GTTTACAATCAGCCACATTCTAGGTAGGGGCCACTTCACCGTACTAACCAGGGAAGCTGTCCCTCACTG
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CCTTAGTACCTAAAGGAAATCTACCCCATCCACACCTGGAGGATTCATCTCTTGTATATGATGAT
CTGGATCCACCAAGACTTGTATGCTCAGGGTCAATTTCTTTTTCTTTTTTTTTTTTTTTTTTTCTTT
TTCTTTGAGACTGGGTCTCGCTTGTGCCCAGGCTGGAGTGGAGTGGCGTGATCTTGGCTTACTGCAGC
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CCTGGGCTCAGGCATCCACCTGTCTCAGCCTCCAGAGTGTGGGATTACAATTGTGAGCCACCACGTC
CAGCTGGAAGGGTCAACATCTTTTACATTCTGCAAGCACATCTGCATTTTACCCACCTTCCCTCTCT
TCTCCCTTTTATATCCATTTTATATCGATCTCTATTTTACAATAAACTTTGCTGCCA
```

Total Sequence Length

Total Length = 2512 bp

Number of A, T, G, and C nucleotides

A	532
T	639
G	621
C	720

GC Content

$$GC = G + C$$

$$GC = 621 + 720 = 1341$$

$$GC\ Content = \frac{1341}{2512} \times 100$$

$$GC\ Content = 53.38\%$$

AT Content

$$AT = A + T$$

$$AT = 532 + 639 = 1171$$

$$AT\ Content = \frac{1171}{2512} \times 100$$

$$AT\ Content = 46.62\%$$

- Total Length: 2512 bp
- A: 532
- T: 639
- G: 621
- C: 720
- GC Content: 53.38%
- AT Content: 46.62%

Q4. Data Interpretation

Two DNA sequences have the following GC contents:

- ☐ Sequence A: 35% GC
- ☐ Sequence B: 68% GC

Answer the following:

1. Which sequence is expected to be more thermally stable?

Sequence B (68% GC)

2. Explain your answer based on the chemical properties of DNA.

Guanine and Cytosine are connected by three hydrogen bonds, while Adenine and Thymine are connected by two hydrogen bonds. Therefore, DNA with higher GC content requires more energy to separate and is more stable at high temperatures.

3. Which sequence is more likely to belong to a thermophilic organism? Justify your answer.

Sequence B (68% GC)

Thermophilic organisms live in high-temperature environments. Their DNA usually contains higher GC content, which provides greater thermal stability and protects the DNA from denaturation.

Q5. Practical Activity

Retrieve one nucleotide sequence from the NCBI Nucleotide database. Submit:

☐ Accession ID ☐ FASTA sequence ☐ Sequence length ☐ GC Content (%) ☐ AT Content (%)

A brief interpretation (100–150 words) explaining what your results indicate about the sequence.

Answer:

Accession ID

JK112949.1

Sequence Name

PP0ADA63YM24FM1 Chlorocebus sabaues PBMCs EST Library Chlorocebus sabaues cDNA, mRNA sequence

GenBank

Sei

PP0ADA63YM24FM1 Chlorocebus sabaues PBMCs EST Library Chlorocebus sabaues cDNA, mRNA sequence

GenBank: JK112949.1

[FASTA](#) [Graphics](#)

Go to: ☑

LOCUS JK112949 341 bp mRNA linear EST 04-FEB-2014
DEFINITION PP0ADA63YM24FM1 Chlorocebus sabaues PBMCs EST Library Chlorocebus sabaues cDNA, mRNA sequence.
ACCESSION JK112949
VERSION JK112949.1
DBLINK BioSample: [SAMN00632079](#)
KEYWORDS EST.
SOURCE Chlorocebus sabaues (Cercopithecus sabaues)
ORGANISM [Chlorocebus sabaues](#)
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Cercopithecidae; Cercopithecinae; Chlorocebus.

FASTA Sequence

PP0ADA63YM24FM1 Chlorocephus sabaeus PBMCs EST Library Chlorocephus sabaeus cDNA, mRNA sequence

GenBank: JK112949.1

[GenBank](#) [Graphics](#)

```
>JK112949.1 PP0ADA63YM24FM1 Chlorocephus sabaeus PBMCs EST Library Chlorocephus sabaeus  
cDNA, mRNA sequence  
GTTTCAGGAGGTGCCAACTGAAGGTATGTGCACTGCGCGTAACCAAGCCGGGCCCGCAGCCGCATCCTCA  
GGGCCGGGGCAAGATCCTCACTTTCGACCAGCTGGCCCTGGACTCCCCAAGGGCTGCGGCACTGTCCT  
GCTCTCTGGTCTCGCAAGGGCCGAGAGGTGTACCGGCATTTTCGGCAAGGCCCCAGGAACCCGCACAGC  
CACACCAAACCTACGTCCGCTCCAAGGGCCGGAAGTTCGAGCGTGCCAGAGGCCGACGGGCCAGCCGAG  
GCTACAAAACCTAACCTTGGATCCTACCCTCTTATTAAGATTTTGAACGCTGACAAA
```

Sequence Length: 341 bp

Nucleotide Count: A: 78 T: 55 G: 94 C: 114

GC Content Calculation

$$GC = G + C = 94 + 114 = 208$$

$$GC\ Content = \frac{208}{341} \times 100$$

$$GC\ Content = 61.00\%$$

AT Content Calculation

$$AT = A + T = 78 + 55 = 133$$

$$AT\ Content = \frac{133}{341} \times 100$$

$$AT\ Content = 39.00\%$$

- Accession ID: JK112949.1
- Sequence Length: 341 bp
- A = 78
- T = 55
- G = 94
- C = 114
- GC Content = 61.00%
- AT Content = 39.00%

Brief Interpretation

The analyzed sequence (JK112949.1) is a cDNA/mRNA sequence from *Chlorocebus sabaeus* (African green monkey). The sequence length is 341 base pairs. Analysis showed a GC content of 61.00% and an AT content of 39.00%. The relatively high GC content indicates that the sequence is likely to be more thermally stable because guanine-cytosine base pairs form three hydrogen bonds, compared to two hydrogen bonds in adenine-thymine pairs. Higher GC content can contribute to increased stability of nucleic acids and may influence gene expression and genome organization. Since this is an expressed sequence tag (EST), it represents a transcribed region of the genome and may be associated with an actively expressed gene. The nucleotide composition provides useful information for sequence comparison, molecular studies, and understanding genetic characteristics of the organism.

Q6. Critical Thinking

Can two DNA sequences have the same length but different GC contents? Explain your answer with a suitable biological example.

Answer:

Yes, two DNA sequences can have the same length but different GC contents.

Example

Sequence 1: ATATATATAT

Length = 10 bp

GC Content = 0%

Sequence 2: GCGCGCGCGC

Length = 10 bp

GC Content = 100%

Although both sequences have the same length, their nucleotide composition is different. As a result, their GC contents and thermal stabilities are also different. The GC-rich sequence is more stable because it contains more GC base pairs with three hydrogen bonds.